

Introduction to Multilevel Measurement Models

SMIP Summer School 2025: Lecture 01

About Your Instructors

- **Lesa Hoffman**: lesa-hoffman@uiowa.edu
- **Jonathan Templin**: jonathan-templin@uiowa.edu

Workshop Introduction

This is an ambitious workshop:

- Measurement models (Item Response Theory)
- Multilevel models for clustered data
- Bayesian statistics
- Multilevel measurement models

Each could be its own workshop

Workshop Materials

Set of slides on topics pulled from other course materials

- In case additional detail on any topic is needed

Full set of analysis files in R and Stan

- In case you would like to work ahead

Workshop Schedule

- Mornings: 9:00-12:30 (9:00-13:00 Friday)
 - Break at 10:45
- Afternoons:
 - Tuesday: 14:00-17:15 (break at 16:00)
 - Wednesday: 15:15-17:45 (break at 16:45)
 - Thursday: 14:00-15:45 (no break)

Lecture and Example Time Plan

Last hour of each day is open

Your choice:

- Questions
- Example practice
- Work with your data
- Leave early

Syntax and Model Note

- As stan models can take a very long time to run, all prior model results have been saved
- Model results are in the `models` folder
- The scriptGuide.pdf file contains the details for each model
<https://www.dropbox.com/s/wvxqd9vha1u8ov5/%21scriptGuide.pdf?dl=0>
- Some slides are written in Quarto (markdown), which can be rendered from the `slides` folder

Workshop Sections

1. Introduction to Measurement Models, Bayesian Statistics, and Stan
2. Introduction to Multilevel Models
3. Introduction to Multilevel Measurement Models

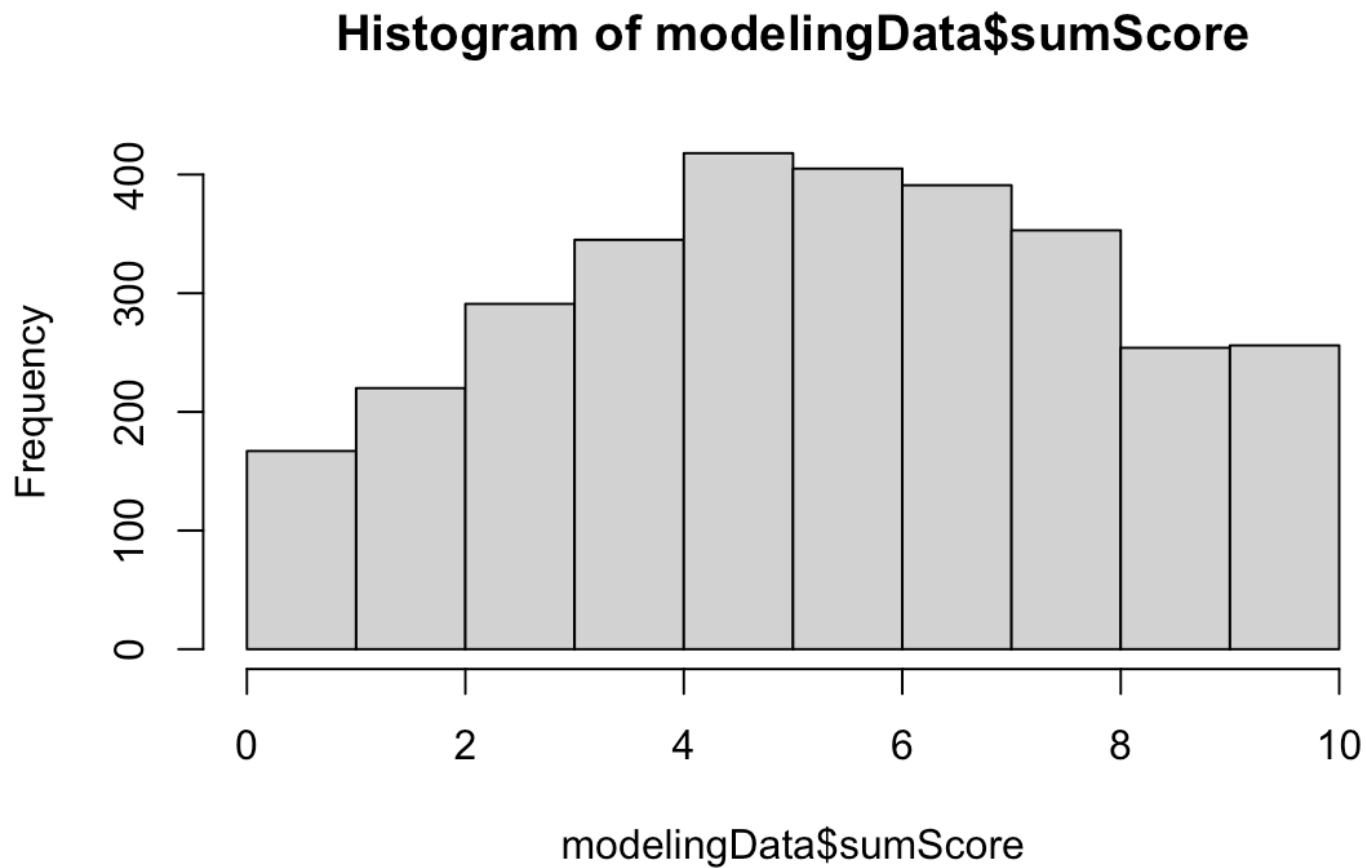
Running Example

Running Example Data

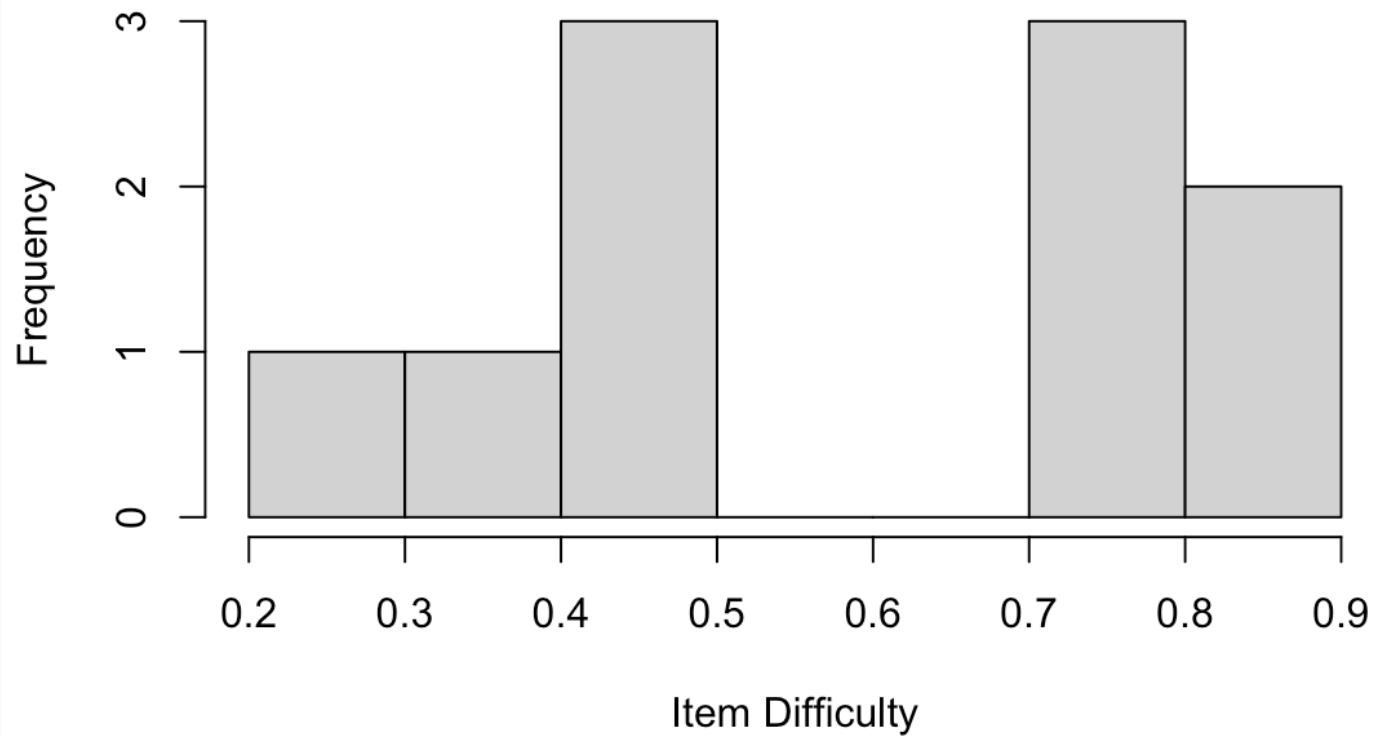
Data come from a secondary school (10th grade) end-of-grade mathematics assessment given around the year 2006 in a midwestern state in the USA

- Sample of 50 students from 62 schools
- Sample of 10 mathematics items from the assessment
 - Items are scored correct/incorrect (`score1` - `score10`)
 - The sum of all 10 items is included (`sumScore`)
- Other variables:
 - `studentID` : Student ID number (created for this example)
 - `schoolID` : School ID number (created for this example)
 - `frlunch` : Free/reduced lunch status (1 = free/reduced lunch, 0 = not free/reduced lunch; indirect indicator of student socioeconomic status)

Distribution of Sum Scores



Distribution of Item Difficulties



Workshop Lectures

- Lecture 01: Introduction to Multilevel Measurement Models Workshop
- Lecture 02: Introduction to Measurement Models
- Lecture 03: Introduction to Bayesian Statistics